

BRADY LIN

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EDUCATION

Northwestern University

B.S. Electrical Engineering | Segal Design Certificate

Evanston, IL
Expected Jun 2028

SKILLS

Embedded: ESP-IDF, Arduino, PlatformIO, STM32, FreeRTOS, embedded Linux, ISRs, I²C, SPI, UART, CAN, RS-485, Modbus

Robotics & Controls: ROS 2, URDF, Pinocchio, bilateral teleoperation, system identification, gravity/friction compensation, PID, DSP

Electronics: KiCAD, EAGLE, LTspice, PCB layout, oscilloscope, logic analyzer, multimeter, JTAG, soldering (THT/SMD)

Systems & Software: C, C++, Python, MATLAB, Bash, JavaScript, Git, CMake, Docker, Linux (Ubuntu)

Mechanical: SolidWorks, Onshape, Fusion 360, GD&T, DFM, CNC, 3D printing, laser/waterjet cutting, machining, most shop tools

EXPERIENCES

Mundane Robotics

Palo Alto, CA

Robotics Hardware & Controls Software Intern

April 2026 – Sept 2026

- Upgraded a bilateral 7-DOF leader–follower exoskeletal teleoperation system from a legacy prototype to next-gen arms, with new URDFs and actuator models, adding 1kHz Pinocchio gravity/friction compensation and ROS 2 communication over Ethernet
- CAD-ed and built a custom motor test bench to QC 40 BLDC actuators across 10 suppliers, saving ~\$30K vs. off-the-shelf rigs
- Modeled viscous/Coulomb friction from motor test-bench data and applied high-frequency dithering to cancel static friction
- Programmed the 3-DOF neck control stack: Pinocchio gravity/friction comp, headset quaternion streaming to RPY setpoints, MATLAB PID tuning, beating the legacy system with 47% less jitter and 120% higher max speed
- Built C++/Python automated test routines interfacing with CAN FD and RS-485 devices, cutting manual testing labor ~75%
- Sourced and negotiated custom actuators/sensors with Chinese and US suppliers in Mandarin, averaging 60% below quote

Formlabs

Somerville, MA

Mechanical Engineering Intern

Dec 2025 – Mar 2026

- Owned development of an embedded GMSL camera and programmable lighting system for a next-gen SLA 3D printer, spanning Onshape chassis design, KiCAD PCB/schematic layout, electromechanical integration, DFM, and C camera driver firmware
- Built an automated imaging rig (2,000+ images across 10 PWM lighting configs, 7 lenses, and 4 sensors) to drive optics selection for the production system, along with a locally hosted dashboard for live stakeholder tuning
- Designed force-sensing fixtures to quantify deflection in resin-mixing arms, logging to CSV via C++ with one-click export to pre-filtered and analyzed Excel reports, driving resin-mixer geometry and material-selection decisions for next-gen prototypes

Center for Robotics and Biosystems

Evanston, IL

Undergraduate Research Engineer

Mar 2025 – Dec 2025

- Architected and built an autonomous battery-swap robot for a 50-drone swarm, developing ESP32 firmware on ESP-IDF for closed-loop stepper control and I²C networking across 20+ nodes
- Reduced X-axis racking error from ~8° to <0.5° through kinematics analysis of the 1.5 m × 1.5 m CoreXY gantry
- Designed 30+ DFM/DFA parts in SolidWorks, fabricated via 3D printing, laser cutting, waterjet cutting, and machining

PROJECTS

Robotic Drummer and Guitarist | Lead Engineer & Founder

Oct 2024 – Present

bradylin.com/projects/pulse/

- Led design and development of a six-armed robotic drummer (mechanical arms, animatronics, LCD feedback) and a robot guitarist on a Linux/Teensy 4.1 system, modeling and validating 60+ mechanical, grounding, and thermal components in CAD
- Developed low-latency C++ firmware coordinating servos and steppers, syncing live MIDI signals via a real-time pre-parser
- Produced a live performance for 100+ attendees, leading cross-team collaboration to showcase the robots playing with a band

Expressive Violin-Style MIDI Instrument | Designer & Hardware Engineer

Nov 2024 – Jul 2025

bradylin.com/projects/stradex/

- Engineered a violin-inspired MIDI controller with an Onshape-modeled enclosure and 3D-printed moving parts, validated across 30+ prototype iterations and >10,000 automated actuation cycles
- Designed and assembled a compact 70+ component KiCAD PCB around an RP2350 with multiple I²C sensors, running C++ firmware (PlatformIO) with DSP and input hysteresis for smooth, intuitive play
- Made a project video earned 16,000+ YouTube views and a Hackaday feature, driving 20 unit sales